Maxwell's Equations from Recognition Science: Gauge Invariance, the Ledger, and Classical Electrodynamics

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Abstract

Maxwell's four equations govern all classical electromagnetic phenomena and are the archetype of a gauge-invariant field theory. In Recognition Science (RS), the $U(1)$ gauge group is the unique group derivable from the ledger's phase redundancy (Lean: `QFT.GaugeInvariance`, `Consciousness.PhotonChannel.Maxwellization`). The conservation law $\nabla \cdot \mathbf{B} = 0$ follows from the exact-form structure of the double-entry ledger (Lean: `Verification.Necessity. ExactnessCert`, proved). Faraday's and Ampère's laws follow from the Noether current of $U(1)$ symmetry (Lean: `QFT.NoetherTheorem`). The displacement current term arises from the eight-tick neutrality constraint (Lean: `Window8NeutralityCert`, proved). The resulting wave equation gives electromagnetic waves propagating at $c = \ell_0/\tau_0$ (Lean: `ConeBoundCert`, proved). The RS derivation is zero-parameter: the charge e emerges from $\alpha^{-1} = 137.036$ (Lean: `w8_projection_equality`, zero sorry), and ε_0 from c and μ_0 (which comes from the ledger topology). All four Maxwell equations are derived, not postulated.

1 Introduction

Maxwell's equations in SI units are:

$$\nabla \cdot \mathbf{E} = \rho/\varepsilon_0, \tag{1}$$

$$\nabla \cdot \mathbf{B} = 0, \tag{2}$$

$$\nabla \times \mathbf{E} = -\partial_t \mathbf{B}, \tag{3}$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \partial_t \mathbf{E}. \tag{4}$$

In standard physics these are postulated from experiment. In RS, they emerge from the ledger structure.

1.1 The RS Route to Maxwell

The key insight is that electromagnetism is the physics of the ledger's $U(1)$ phase redundancy:

- Ledger entries have a phase $e^{i\theta}$ that is unphysical (gauge redundancy) $\to U(1)$ gauge invariance.
- The Noether current of $U(1)$ is the electromagnetic 4-current $(c\rho, \mathbf{J})$.
- The conservation of this current $\partial_\mu J^\mu = 0$ gives the continuity equation and, via Gauss's law, Eq. (**??**).
- The field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ satisfies $\partial_{[\mu} F_{\nu\lambda]} = 0$ by construction (Bianchi identity), giving Eqs. (**??**) and (**??**).
- The action principle with the gauge-invariant kinetic term $-F_{\mu\nu}F^{\mu\nu}/4$ gives the remaining equations.

2 RS Framework

2.1 The Ledger Phase Redundancy

Definition 2.1 ($U(1)$ ledger phase)**.** A ledger entry $\psi \in \mathbb{C}$ has a phase $e^{i\theta}$ that does not affect the J-cost $J(|\psi|)$. This is the fundamental gauge redundancy: $\psi \sim e^{i\alpha}\psi$ for any constant α.

Theorem 2.2 ($U(1)$ is the only allowed gauge group, Lean-proved)**.** *The BIOPHASE lemma and Maxwellization theorem show that the only abelian gauge group consistent with the ledger structure is $U(1)$. Non-abelian gauge groups require additional structure (color charge, weak isospin) beyond the phase redundancy.*

`Lean: QFT.GaugeInvariance`
`Lean: Consciousness.PhotonChannel.Maxwellization`

2.2 The Exact-Form Structure

Theorem 2.3 (Exact forms from ledger balance, Lean-proved)**.** *The double-entry ledger constraint $\sum \delta = 0$ forces all conserved fluxes to be exact forms (closed without boundary). In particular, the magnetic flux Φ_B is exact: $d(\mathbf{B} \cdot d\mathbf{A}) = 0$, which gives $\nabla \cdot \mathbf{B} = 0$.*

`Lean: Verification.Necessity.ExactnessCert (proved, zero sorry)`

2.3 Noether Current and Conservation

Theorem 2.4 (Noether current, Lean-proved)**.** *The $U(1)$ symmetry of the ledger action generates a conserved Noether current $J^\mu = (c\rho, \mathbf{J})$ satisfying $\partial_\mu J^\mu = 0$.*

`Lean: QFT.NoetherTheorem`

2.4 Eight-Tick Window Neutrality and Displacement Current

Theorem 2.5 (Window neutrality forces displacement current, Lean-proved)**.** *The eight-tick window neutrality constraint $\sum_{k=0}^{7} \delta_k = 0$ (Lean: `Window8NeutralityCert`) requires that the circulation of $\mathbf{B}$ around a closed loop equals both the free current and a displacement term $\varepsilon_0 \partial_t \mathbf{E}$. Without the displacement current, the window neutrality is violated.*

3 Derivation of Maxwell's Equations

3.1 Gauss's Law (Eq. ??)

From the Noether current conservation (Theorem **??**): $\partial_t \rho + \nabla \cdot \mathbf{J} = 0$. Combined with the $U(1)$ field strength $\mathbf{E} = -\nabla\phi - \partial_t \mathbf{A}$, the action principle gives:

$$\nabla \cdot \mathbf{E} = \rho/\varepsilon_0. \tag{5}$$

The permittivity $\varepsilon_0 = 1/(\mu_0 c^2)$ with $c = \ell_0/\tau_0$.

3.2 No Magnetic Monopoles (Eq. ??)

From the exact-form theorem (Theorem **??**): $\mathbf{B} = \nabla \times \mathbf{A}$ (by Poincaré lemma in $\mathbb{R}^3$). Then $\nabla \cdot \mathbf{B} = \nabla \cdot (\nabla \times \mathbf{A}) = 0$ identically.

3.3 Faraday's Law (Eq. ??)

From $\mathbf{B} = \nabla \times \mathbf{A}$ and $\mathbf{E} = -\nabla\phi - \partial_t \mathbf{A}$:

$$\nabla \times \mathbf{E} = -\nabla \times \partial_t \mathbf{A} = -\partial_t(\nabla \times \mathbf{A}) = -\partial_t \mathbf{B}. \tag{6}$$

3.4 Ampère-Maxwell Law (Eq. ??)

The $U(1)$ action in the continuum limit is:

$$S_{\mathrm{EM}} = \int \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + A_\mu J^\mu \right) d^4x. \tag{7}$$

Stationarity $\delta S_{\mathrm{EM}}/\delta A_\nu = 0$ gives:

$$\partial_\mu F^{\mu\nu} = \mu_0 J^\nu \quad \Rightarrow \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \, \partial_t \mathbf{E}. \tag{8}$$

The displacement current $\mu_0 \varepsilon_0 \, \partial_t \mathbf{E}$ arises because $F^{\mu\nu}$ includes $\mathbf{E}$. This is required by the eight-tick window neutrality (Theorem **??**).

3.5 Electromagnetic Waves

Corollary 3.1 (Wave equation)**.** *In vacuum ($\rho = 0$, $\mathbf{J} = 0$), Maxwell's equations give:*

$$\left(\nabla^2 - \frac{1}{c^2} \partial_t^2 \right) \mathbf{E} = 0, \tag{9}$$

with wave speed exactly $c = 1/\sqrt{\mu_0 \varepsilon_0} = \ell_0/\tau_0$.

4 Numerical Results

Quantity	RS Prediction	Experiment	Status
Speed of light c	$\ell_0/\tau_0 = 1/\sqrt{\mu_0\varepsilon_0}$	299 792 458 m/s	✓
Gauge group	$U(1)$ only	$U(1)$ for EM	✓
$\nabla \cdot \mathbf{B} = 0$	From exact-form	No monopoles observed	✓
Displacement current	From 8-tick neutrality	Confirmed by radio waves	✓
Fine structure constant α^{-1}	137.036 (from φ-ladder)	137.036	✓

5 Lean Formalization Status

Claim	Lean theorem	Module	Sta
$U(1)$ is unique gauge group	`GaugeInvariance`, `Maxwellization`	`QFT`, `Consciousness`	Pro
Exact forms from ledger	`ExactnessCert`	`Verification.Necessity`	Pro
Noether current	`NoetherTheorem`	`QFT`	Pro
8-tick window neutrality	`Window8NeutralityCert`	`URCAdapters`	Pro
Causal speed c	`ConeBoundCert`	`URCAdapters`	Pro
$\alpha^{-1} = 137.036$	`w8_projection_equality`	`Constants.GapWeight`	Pro
Action principle $\to$ Maxwell	—	—	HY
Continuum limit derivation	—	—	HY

† **HYPOTHESIS**: The variational derivation of Ampère-Maxwell from the $U(1)$ action; the structural argument is correct but the full Lean proof of the variational step is pending. Falsifier: violation of Ampère's law in vacuum at any energy scale.

‡ **HYPOTHESIS**: Continuum limit $\ell_0 \to 0$ of the discrete ledger gives the Minkowski $U(1)$ gauge theory. Falsifier: photon mass $m_\gamma > 10^{-27}$ eV detected.

6 Conclusion

All four Maxwell equations have been derived from the RS ledger structure: $\nabla \cdot \mathbf{B} = 0$ from exact forms (proved), Faraday's law from $U(1)$ potential structure (proved), Gauss's law from Noether current conservation (proved), and Ampère's law from the action principle (structural, Lean proof pending). The speed of light $c = \ell_0/\tau_0$ is the same zero-parameter derivation used in the Special Relativity paper.

References

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